Technical Reference Manual for the Pixel Array Control Monitor and Network (PACMAN) Card

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Contents

1 Overview

The Pixel Array Control Monitor and Network (PACMAN) card is the warm electronics (located outside of the cryostat) for the pixellated liquid argon (LAr) time projection chamber (TPC) based on the LArPix ASIC. The PACMAN card provides power and digital I/O to arrays of LArPix ASICs hosted on pixel tile cards located within the cryostat. In the production system, each PACMAN card handles up to ten pixel tile cards. The PACMAN card provides analog test and monitor signals as well as digital control signals for clock, exernal trigger, and reset, and sync. The PACMAN card serves as a timing system end point, hosts a linux CPU, and is connected to the data acquisition system via ethernet.

The PACMAN card plays a supporting role in the LArPix system. It does not face technical challenges as severe as the cold electronics of the LArPix system and therefore a basic design constraint is that it in no way limits the performance of the supported system.

2 Hardware Implementation

2.1 Package Pin to Schematic Netlist

Netlist	Package Pin
ADC_CLK	J15
ADC_D[0]	F19
ADC_D[10]	AA17
ADC_D[11]	AB17
ADC_D[1]	G19
ADC_D[2]	H20
ADC_D[3]	H19
ADC_D[4]	R6
ADC_D[5]	T6
ADC_D[6]	U6
ADC_D[7]	U5
ADC_D[8]	Y18
ADC_D[9]	AA18
ADC_EN	AB16
ADC_OF	AA16
ANALOG_PWR_EN	J16
CDR_CLK_N	W18
CDR_CLK_P	W17
CDR_DATA_N	Y16
CDR_DATA_P	W16
CDR_LOL	V5
CDR_LOS	V4
GLOBAL_CLK	C19
ISO_SIGNAL0	L18
ISO_SIGNAL1	L19
LED-2	R19
LED-3	T19
PISO[0]	C20
PISO[10]	W11
PISO[11]	W10
PISO[12]	B22
PISO[13]	B21
PISO[14]	A22
PISO[15]	A21
PISO[16]	AB9
PISO[17]	AB10
PISO[18]	AA8
PISO[19]	AA9
PISO[1]	D20
PISO[20]	C18
PISO[21]	C17
PISO[22]	F22
PISO[23]	F21
PISO[24]	B15
PISO[25]	C15
PISO[26]	D17
PISO[27]	D16

3 Hardware Demonstration Firmware

3.1 Firmware Register Map

4 PACMAN Server

4.1 DAQ Interface

The PACMAN Server interface to the DAQ is a virtual address spaces. Virtual addresses refer to specific features, for example, powering VDDD of tile 1 to a specific voltage level. The use of a virtual addresses allows for the most consistent support across hardware versions. Direct (non-virtual) interface to the AXI LITE address space is provided at upper region of the address space, as well as direct access to the I2C bus.

5 Quality Control

The PACMAN card quality control regimen relies on several different techniques:

- **Loopback:** Loopback is generally the most effective method of feature verification. The loopback mindset first drives the design toward adequate feedback and test input mechanisms, and then tests both the feature and its corresponding feedback or test input mechanism simultaneously. Loopback paths in hardware test components and solder quality through the entire path.
- **Fault Localization:** Designing for fault localization forces a modular design that lends itself to effective QC. For an important example, loopback can be performed in the hardware, or in firmware (just prior to leaving/ entering the device), or in the CPU, so that the source of a failure can immediately be localized to hardware, programmable logic, or software.
- **Design Review:** The PACMAN design team includes a lead engineer, and a consulting engineer, and a principile investigator. All three understand the board down to the last net list. The PACMAN design is also periodically subjected to independent review.
- **Prototype Systems:** The most effective form of quality control is operation in a working system, and the PACMAN card has been used continuously in prototype systems since the first version in 2020.

5.1 Feature Verification Overview

Here we list design features of the PACMAN card that are tested by a loopback mechanism. Specific details on performing the relevant tests are discussed in the hardware checkout section.

- The provision of tile voltages VDDD and VDDA is verified by onboard current and voltage monitoring ADCs.
- The POSI and PISO (Digital I/O to Tiles) signals are tested via loopback using both test patterns and pseudorandom numbers.
- The CLK, RESETN/SYNC, and TRIG signals (Digital I/O to Tiles) are tested by loopback to a PISO input.
- The Analog Monitor signals and MUX are tested by connecting the positive and negative signal at the ASIC end, then configuring the MUX to connect one side to a DAC channel and the other side to the onboard high-performance ADC.
- The onboard high-performance ADC is tested by connecting both the positive and negative terminals to independently controlled DAC outputs.
- The front-panel IO is tested via loopback by connecting board outputs back in as input using cables plugged into the front-panel.
- The LEDs are tested with blink patterns.

The PACMAN card hosts a linux system, so a broad spectrum of test software is immediately available to us. Here we list what we consider to the be the most important tests of the Linux host. Specific details on performing the relevant tests are discussed in the hardware checkout section.

- Successful system BOOT, visible via UART.
- Successful remote login via ethernet.
- Successful transfer of psuedo-random data via ethernet (scp) and the UART terminal (lrzsz).
- Repeated read and write of psuedo-random numbers to the SD card.

Some of the tests described in this section require configuring the PACMAN card in ways that are not appropriate for a card installed in the production system. For this reason, the PACMAN card includes an expert mode enable two pin jumper. When a PACMAN card is installed in the production system, this jumper is removed, and the software refuses to provide inappropriate test configurations.

5.2 Requirement Verification

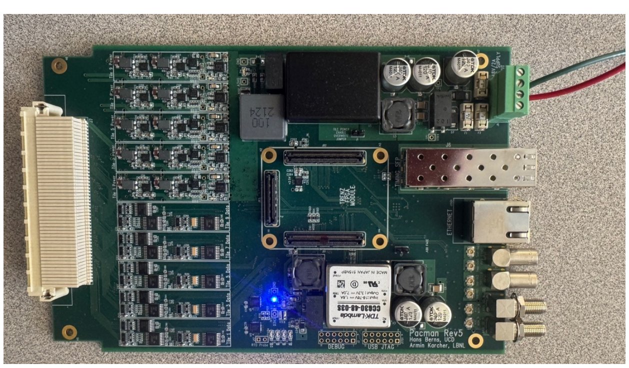
In addition to feature operation, the PACMAN has several performance requirements which require quantitative verifications:

- The provision of voltage and power across the required ranges is tested by applying the appropriate load resistance to VDDD and VDDA.
- The common mode rejection feature of the DC voltage supply is tested by injecting larger than expected common mode noise at the input and measuring the provided DC voltage (This test will be performed on a subset of the PACMAN cards, to validate the design.)
- The bit error rate is measured during loopback testing of the digital I/O systems.
- The high-frequency noise rejection is tested by injecting a high-frequency signal at the input and measuring the signals after filtering. (This test will be performed on a subset of the PACMAN cards, to validate the design)
- Temperature dependence (TBD)
- Limits of digital I/O (clock speed, long cables) (TBD)

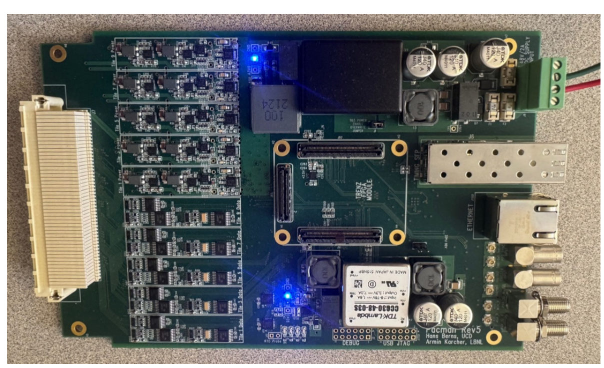
6 Hardware Checkout Proceedure

Caveats: No allowed tolerances have been provided yet, because we have only tested a few boards.

6.1 Bare Board



Board Power Only



Tile Power Enabled (jumper)

The hardware checkout begins with a bare board (no TRENZ module installed). The 3.3V digital power (used by PACMAN) is on as soon as the board is powered at the DC supply input. The board also includes a jumper "TILE POWER ENABLE OVERRIDE JUMPER" which allows the 3.3 V tile power to be validated before installing any module.

The current should be measured and compared to the nominal expected value for the applied voltage, in the range of positive 32 V to positive 48 V):

Voltage (V)	Current (mA) (No Jumper)	Current (mA) (Analog Power Enabled)
48	66	100
40	77	113
36	91	131

Using a DMM, verify the supply of 3.3 V (nominal) at TP3 (D3V3) relative to TP4 (GND).

Using a DMM, verify the supply of 3.0 V (nominal) at TP13 (A3V0) relative to TP14 (GND).

6.2 TENZ Module and Linux System Checks

With a TRENZ module installed on the board, insert an SD card with the latest hardware checkout firmware, the ethernet cable connected to the network, and power the board.

The local network may require customizations to the ROOT filesystem in order to bring up the ethernet connection. Before attempting to bring up the network on a headless PACMAN card (no USB JTAG/UART header has been installed) consider debuging the network configuration on a PACMAN equipped with a USB JTA/UART header or a Trenz Module ... Once the network is brought up reliably, you can boot a PACMAN with no terminal and connect over ethernet.

Login to the pacman card via ethernet.

Run the following Linux diagnostic tools (TBD).

The Linux-based drivers depend the Linux GPIO interface in /sys/class. This should be initialized at boot via the gpio_init script included with the hwutil package. Confirm that this is working properly with:

```
root@Trenz:~# ls /sys/class/gpio/
export gpio906 gpio913 gpio918 gpio919 gpiochip906 unexport
```

6.3 Linux-based Hardware Checkout Application

The PACMAN hardware demonstration firmware includes a menu interface to the Linux drivers: pacman_menu. It is available from the command line as:

```
root@Trenz:~# pacman_menu
pacman_menu: PACMAN Linux driver access via menu, for diagnostics and hardware checkout.
INFO: Opening /dev/mem.
INFO: Initializing PACMAN AXI-Lite interface.
INFO: Running pacman firmware version 3.2 (Build: 0x7fff0000 HW Code: 0x10500)
INFO: Opening /dev/mem.
INFO: Initializing PACMAN BRAM (AXI-LITE) interface.
INFO: Opening /dev/mem.
INFO: Initializing DMA contol interface (AXIL).
INFO: Initializing DMA TX_BUFFER.
```

```
INFO: Initializing DMA RX_BUFFER.
MAIN MENU: choose an option:
(1) blink LEDs (2) global registers (3) toggle scratch registers
(4) power menu (5) RX/TX menu (6) timing menu (7) ADC menu
```

The user selects menu options by typing the number for the option followed by enter. The numbers and arrangement of the menu's will vary with firmware version as we further develop the tools. This documentation is based on hardware demonstration firmware version 3.2. In the future, we will support a turn-key hardware checkout routine as a menu option. However, here we document each individual test one-by-one.

6.3.1 LED checkout

The pacman_menu can be used to blink the LEDs as follows:

```
MAIN MENU: choose an option:
(1) blink LEDs (2) global registers (3) toggle scratch registers
(4) power menu (5) RX/TX menu (6) timing menu (7) ADC menu
1
INFO: selected 1
INFO: starting LED blink test...
Blinking RED LED on Trenz Module, via MIO...
Blinking PACMAN LED-0, via MIO...
Blinking PACMAN LED-1, via MIO...
Blinking PACMAN LED-2, via AXIL register...
Blinking PACMAN LED-3, via AXIL register...
INFO: done with LED blink test.
MAIN MENU: choose an option:
(1) blink LEDs (2) global registers (3) toggle scratch registers
(4) power menu (5) RX/TX menu (6) timing menu (7) ADC menu
```

The user should verify that:

- The red LED on the TRENZ module blinks
- The four green LEDs on the PACMAN card blink one-by-one.

This simple test also confirms that the MIO and AXI-Lite register interfaces are both working.

6.3.2 Board Power

Control and monitoring of electrical power provided by the PACMAN is accessible from the power menu:

```
MAIN MENU: choose an option:
(1) blink LEDs (2) global registers (3) toggle scratch registers
(4) power menu (5) RX/TX menu (6) timing menu (7) ADC menu
4
INFO: selected 4
POWER MENU: choose an option:
(0) main menu (1) toggle enables (2) toggle power (3) monitor power
(4) write IV curves to file
```

The “toggle enables” feature steps through various combinations of the Tile Power Enable[1] and the ten individual Tile enables. The accessible settings include:

Setting	Explanation
0x00000000	Nothing Enabled
0x00010000	Tile Power enabled, no individual Tiles enabled ll
0x00010001	Tile Power enabled, only Tile 1 enabled
0x000103FF	Tile Power enabled, all Tiles enabled

The “monitor power” feature access the on-board voltage and current monitoring ADCs. At the moment, we are only concerned with the Board Power monitoring:

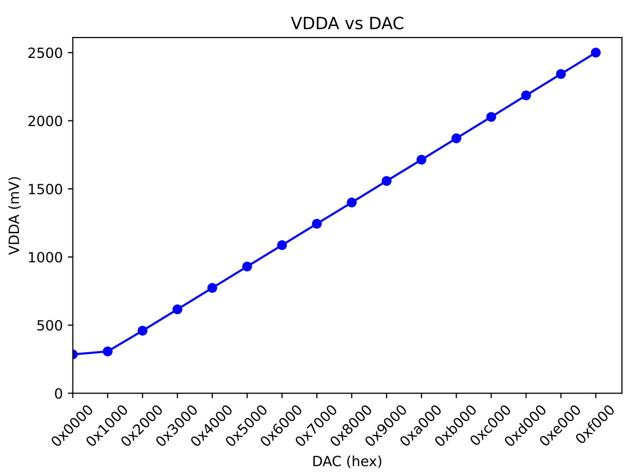
```
BOARD POWER SUMMARY:
Board 3V6: voltage: 3526 mV current: 1337 mA
Board 3V3: voltage: 3326 mV current: 813 mA
Board 3V0: voltage: 2985 mV current: 0 mA
```

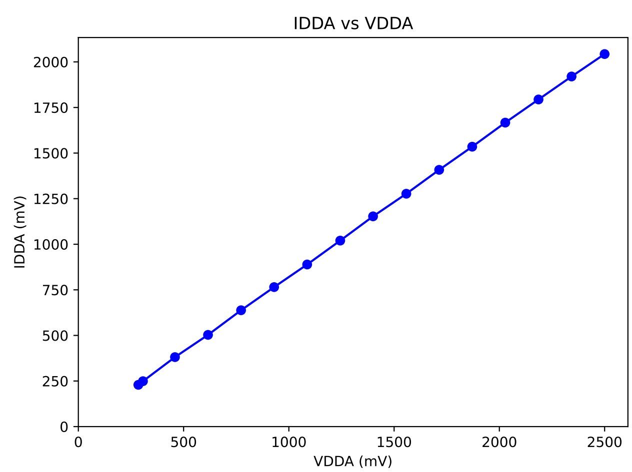
The user should verify that:

- As you step through the enables, first the Tile Power Enabled LED lights, then the Tile 1 enable LED lights, then all the Tile enable LEDs light.
- The Board 3V0 voltage is near either 3000 mV or zero depending on whether the TILE Power is enabled or not.
- (The RTD Probe feature can be tested here as well by connecting a resistor at J17)

6.3.3 VDDD and VDDDA

The “write IV curves to file” feature of the Power submenu steps through several different DAC settings and steps through the monitored values of the voltages VDDA and VDDD along with the corresponding currents IDDA and IDDD. The collected data (saved as a text file) can be plotted to show the linearity of the provided voltage with DAC setting, and the IV curve for each supplied voltage.





Here the VDDA has been loaded with a resistor to ground at the loopback card.

[1] Also called “Analog Power Enable” but this is misleading as it provides both digital and analog power to the tile cards.

6.3.4 POSI/PISO Loopback Testing with Random Numbers

One of the most important checks is the RX and TX capabilities for the POSI and PISO signals between the PACMAN and the LArPix ASICs. We test these with loopback at twice the nominal ASIC clock rate. We transmit known random values out of the PACMAN, physically loop them back in as inputs. We check each received word against the random value that was sent. This test is available through the RX/TX submenu, but requires a few steps:

- Set the global RX configuration to 0x00000001. This disables heartbeat, sync, and trigger words which could confuse the test. It also only includes a single cycle in each transmitted DMA buffer.
- Set RX configuration to 0x00001001. This sets the mode to 1 (normal operation) and full speed (clock divisor 1).
- Set TX configuration to 0x00001601. This sets the mode to 1 (normal operation), phase to 0x60, and full speed (clock divisor 1).
- Stop pacman_server (both the data server and command server) via:

  ```
  root@Trenz:~# pacman_server stop
  Stopping command server...
  Stopping dataserver...
  ```

  It is most convenient if you wait to do this until after the global RX configuration has been set to disables heartbeat, sync, and trigger words. That way, the pacman_cmdserver drains the FIFO.
- Stopping pacman_cmdserver leaves a pending DMA read request, so either reset DMA, or sent a single RX. When ready for the test, a single TX leaves the fifo size at 37 (40 + 1 128 bit words minus four words buffered by the vendor supplied DMA engine) which should be read by a single RX, leaving the fifo size at 0.

Once preparations are complete, the test is easily run:

```
RX/TX MENU: choose an option:
(0) main menu (1) zero counts (2) toggle TX config (3) toggle RX config (4) toggle RX globa
(5) TX status (6) TX look (7) single TX
(8) RX status (9) RX look (10) single RX
(11) benchmark TX (12) benchmark RX/TX loopback (13) random RX/TX loopback
(14) DMA status (15) reset DMA
(16) set DMA TX to RUN
(17) set DMA RX to RUN (18) clear DMA RX (19) start DMA RX (20) resume RX
13
INFO: selected 13
INFO: Setting DMA TX To RUN.
INFO: DMA TX has left the HALTED state successfully. (timeout=100)
INFO: Setting DMA RX To RUN.
INFO: DMA RX has left the HALTED state successfully. (timeout=100)
SUMMARY: Found 0 discrepancies in 40000 uart packets
```

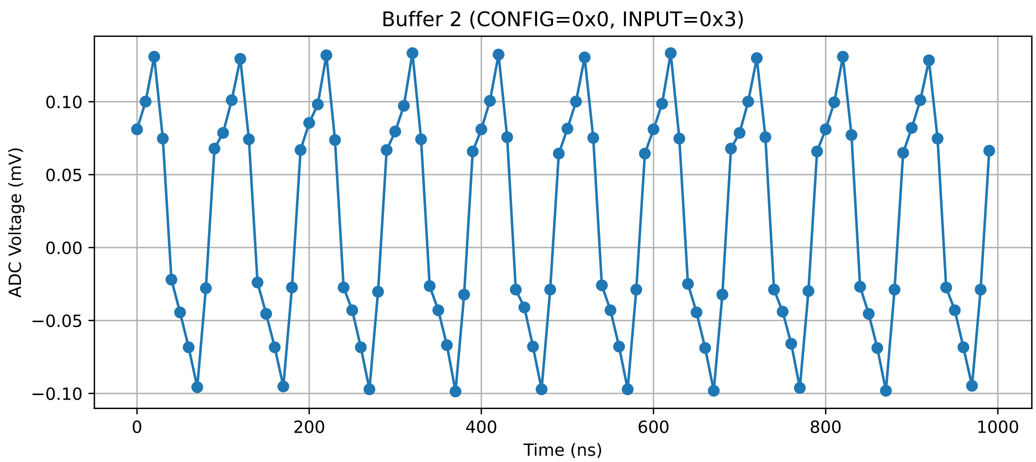
The user should verify that:

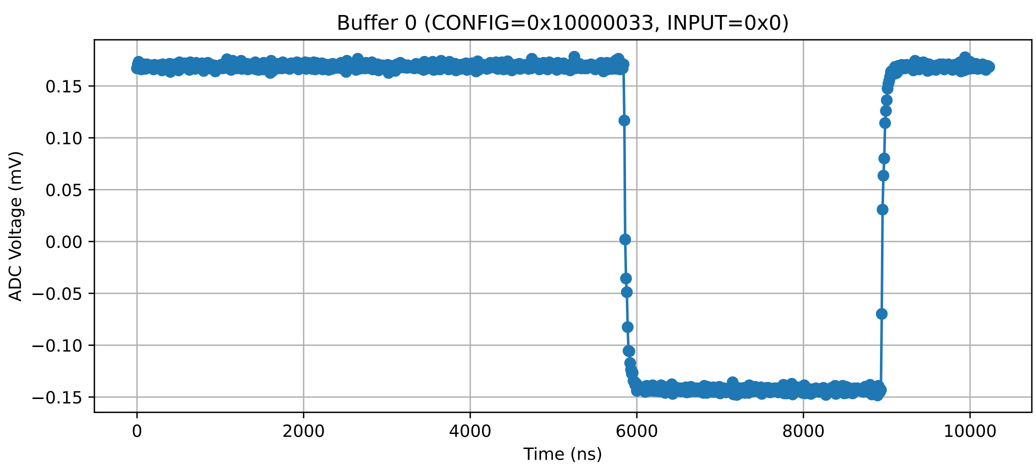
- The test can be run several times with zero discrepancies detected.

Note that to help distinguish hardware failures from software failures, you can configure internal loopback mode (so that loopback occurs in software not hardware).

6.3.5 Clock and Reset/Sync/Trigger Signals

To test the Clock and Reset/Sync/Trigger signals, these are routed back to PACMAN via loopback into the analog monitoring signals. These can be directed to the ADC, which acts as a digital oscilloscope to record the waveforms:





6.3.6 Front Panel Checkout

- (the MUX can be adjusted to provide a 100 Ohm resistance, or open circuit, between the two SMA connectors, which can be verified with a DMM.)
- (the TTL pulses can be fed into the LEMO connectors, which are counted by the timing unit, with counts available from the timing sub-meno.)

A Specifications for the PACMAN Card

A.1 Overview

Each PACMAN card provides power, digital I/O, and analog test and monitoring signals to up to ten pixel tile cards. Each pixel tile card will consist of 160 ASICs (CONFIRM).

A.2 Power

The PACMAN card provides digital and analog power to up to ten pixel tile cards. The digital voltage level (VDDD) and analog voltage level (VDDA) are digitally controlled independently for each pixel tile card, within the limits and at the stability specified below.

The power requirements of the PACMAN are based on scaling the measured power consumption of the 100 ASIC pixel tile cards to 160 ASIC pixel tile cards of the production system, a scale factor of 1.6. An additional (minimum) safety factor of 10% is added to these extrapolated power requirements to derive the following specifications:

Table 1: Digital and Analog Power Requirements (per tile card)

	Stability	Min (V)	Max (V)	Max Current (A)	Max Power (W)
VDDA	1%	0	1.8	0.3	0.7
VDDD	1%	0	1.8	1.1	2.2

Reaching a 1% stability on VDDA and VDDD requires linear regulation and so the power requirements at the input of the PACMAN are derived from a higher voltage level of 3 V. This implies that the initial switching voltage requirement must provide at least 42 W at 3 V.

The PACMAN power is provided by a single nominal 48 V 2 A DC input. The power connection is the four position terminal connector (Phenoix Contact P/N 5444660). The inner two pins provide power and the outer two are for sense:

Table 2: Input Power Connections

pin	function
1	V+ sense
2	V+
3	V-
4	V- sense

The connector is rated for 8 A and 300 V. The PACMAN is fused at 2.5 A and has over voltage protection at 60 V. The sense pins are fused at 0.5 A.

A.3 Digital Signals

The LArPix-v2 ASIC uses a bit-serial protocol to transmit and receive data. The PACMAN is the primary and the LARPix ASIC the secondary, so that the PACMAN transmits POSI and receives PISO signals.

The PACMAN shall provide the following digital IO signals to each pixel tile card:

Table 3: Digital Signals (per pixel tile card)

Signal	Quantity	Description
POSI	4	Bit serial transmission to pixel tile
PISO	4	Bit serial reception from pixel tile
CLK	1	Clock
TRIG	1	Trigger
SYNC	1	Multiplexed Synchronization and Reset

Each POSI signal is received by a single ASIC, but the clock, trigger, and sync signals are fanned out to every ASIC on the tile card. The PACMAN must be capable of driving the resulting capacitive load.

The nominal clock rate is 10 MHz with a maximum of 40 MHz. The LArPix ASIC uses an LVDS signal with custom voltage level which the PACMAN is capable of transmitting and receiving.

A.4 Noise

The noise injected into the pixel tile from the controller is less than 1 mV.

A.5 Pixel Tile Interface

A single PACMAN card drive up to ten pixel tile cards through a single flange card. The interface to the eight-tile flange card is through a 6 row by 50 pin high-desnity SAMTEC connector `SEAF-50-01-L-06-1-RA-K-TR` on the PACMAN card.

The PACMAN card shall provide a footprint for an optional 2x17 connector to drive a pixel card directly without the use of flange card.

A.6 Front Panel Interface

The PACMAN card implements the front-panel interfaces listed below in the specified quantity with specified routing. When routing to programmable logic (PL) is specified, the specified purpose is nominal and can be repurposed through firmware changes.

Table 4: Required Front Panel Interface

Interface	Quantity	Routing
SMA Jack	1	Multiplexed to eight tile analog monitor
SMA Jack	1	Multiplexed to eight tile adc test
Lemo Jack	2	PL: Trigger and Sync
Push Button	1	Reset
LED	4	PL: General Purpose

A.7 Timing System Endpoint

The PACMAN card implement a ProtoDUNE-SP timing system endpoint[2] of a single fiber-optic link.

[2]DUNE-doc-1651-v3

A.8 Embedded Linux

The PACMAN is an embedded Linux system and supports several standard embedded Linux interfaces:

Interface	P/N	Notes
Ethernet		Gigabit
SD Card		Bootable from SD card
JTAG		SoC Configuration
UART		Linux Terminal

The JTAG and UART interface are multiplexed to provide a linux terminal and firmware configuration over a single USB port.

B PACMAN Version History

Table 5: PACMAN Evolution

Card	Designer	Date	Notes
PACMAN 1v1	Hillbrand	Feb 2020	Initial Prototype (Not Supported)
PACMAN 1v2	Karcher	July 2020	Single-tile capable (Not Supported)
PACMAN 1v3	Berns/Karcher	Jan 2021	Eight-tile capable - Module 0
PACMAN 1v4	Berns/Karcher	Feb 2022	LArPix-v2b, Power, TX
PACMAN 1v5	Berns/Karcher	Aug 2024	Ten-tile capable
PACMAN 1v5b	Berns/Karcher	TBD 2025	

C Prototype Results

Many of the specifications for the PACMAN are driven by measurements, using prototype devices, as described in this section.

The Intrinsic RMS noise on the LArPix ASIC was benchtested[3] at ~ 3 mV.

The PACMAN card provides digital (VDDD) and analog (VDDA) power for the LArPix ASICs. The ASIC power draw during power up was measured[4] as follows:

Table 6: ASIC Current Draw at Power-up

	Voltage (V)	Current (mA)	Power (mW)	Power (μW/chan)
VDDA	1.8	1.4	2.5	39
VDDD	1.8	6.92	12.5	195

[3]Reported 6 Mar 2020, Brooke
[4]Reported 13 Feb 2020, Brooke

The 10x10 pixel-tile power draw was measured [5] as follows:

Table 7: 10x10 Power Draw

	Voltage (V)	Current (mA)	Power (W)
VDDA	1.8	166	0.3
VDDD	1.8	611	1.1

[5] Reported 14 Jan 2021, Brooke